

ARNAB KUMAR DEY

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EDUCATION

Bachelor of Technology (BTech) in Artificial Intelligence
Alipurduar Government Engineering and Management College
Minor Degree in Artificial Intelligence
Indian Institute of Technology (IIT), Raipur (RPR)

Expected Graduation: 2028
YGPA: 8.07
Completed
Project Grade: 9.1

RESEARCH EXPERIENCE

Research Intern – RAG / AI Systems

Institute of Engineering & Management (IEM), Kolkata

Present

Advisor: Dr. Subhabrata Banerjee

- Developing **RAG-ISRO**, a secure multi-agent RAG system for aerospace knowledge retrieval in air-gapped environments with zero-trust security, dynamic access control, and evidence verification.

Research Intern – Image Processing / Medical AI

Jadavpur University

Research Project

- Worked on **multimodal medical image analysis** using vision models, medical RAG, evidence verification, agentic reasoning, and uncertainty-aware responses.

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, C, C++, Dart

Frontend: React.js, Flutter, HTML, CSS, Tailwind, Bootstrap, Next.js

Backend & Databases: Node.js, Express.js, MySQL, Firebase, Inngest, Supabase

AI/ML: PyTorch, Neural Networks, NLP, RAG, LLMs, Computer Vision, OpenCV, XAI

Tools & Platforms: Git, GitHub, Vercel, Render, Docker, Google Cloud, Jupyter

PROJECTS

- RAG-ISRO:** Secure multi-agent RAG engine for aerospace technical specifications and compliance knowledge in air-gapped environments with zero-trust security and evidence verification.
- PassonRide:** Flutter-based P2P vehicle rental and AI-powered guided tour marketplace connecting vehicle owners, hosts, and travelers.
- Real-Time Work Detection:** Offline AI assistant designed to track, guide, and validate procedural experiments inside science modules of the Bharatiya Antariksh Station.
- Business Challenges – AI Solutions:** Developed end-to-end AI solutions for real-world industry bottlenecks.
- HealthWise:** ML-based symptom detection system for preliminary health analysis.
- OpenCV Projects:** Computer-vision applications including an interactive AI story generator.
- Roof-Top-Calc:** Computer-vision utility for automated rooftop surface calculation.
- LLM Bias Mitigation:** Tool for detecting, analyzing, and mitigating algorithmic bias in LLMs.

COMPETITIVE PROGRAMMING & ACHIEVEMENTS

Platforms: LeetCode, NeetCode, GeeksforGeeks (GFG) **Core:** Dynamic Programming, Hash Tables, Greedy Algorithms, Arrays, Strings, Data Structures.

ADDITIONAL INFORMATION

Languages: English (Fluent), Hindi (Fluent), Bengali (Native)

Currently Learning: Agentic AI, Advanced Machine Learning, Cloud Computing (AWS/Azure), DevOps Technologies

Summary: Aspiring Software & AI Engineer experienced in AI/ML, RAG, computer vision, full-stack development, and research-oriented systems.